

SUMMARY

Generative AI engineer with 3 years shipping LLM-powered products end to end: production RAG pipelines, LangChain and LangGraph agents, multi-tenant SaaS on Python and FastAPI. Strong on retrieval quality – hybrid search with reciprocal rank fusion, RAGAS and precision / recall / MRR evals gated in CI. Guardrails cover prompt-injection detection and PII redaction.

TECHNICAL SKILLS

Generative AI & Agentic Systems: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents & Tool Use, Agentic Workflows, Tool Calling, Structured Outputs, LangChain (LCEL), LangGraph, Prompt Engineering, Prompt Caching, Embeddings, Hybrid Search (RRF)
AI Evaluation & Responsible AI: RAGAS, Precision / Recall / MRR, CI eval harness, Prompt-injection defense, PII redaction
Languages: Python, TypeScript, JavaScript, SQL
Python & Backend: FastAPI, Pydantic, asyncio, Uvicorn, SQLAlchemy, Alembic, REST / OpenAPI, JWT / OAuth2, SSE, Node.js, NestJS
Data & Vector Stores: pgvector (Vector Database), PostgreSQL (RLS, Full-Text Search), DuckDB, MongoDB (NoSQL), Redis
Cloud, DevOps & Frontend: AWS (EC2, S3, RDS), Docker, Docker Compose, GitHub Actions (CI/CD), Git, Linux, React, Next.js

EXPERIENCE

Full Stack Developer

Aug 2025–Present

A&J Intelli Systems Pvt Ltd

Gurugram, India (Remote)

- Designed and shipped a production RAG pipeline end to end: ingestion, chunking, embeddings, pgvector semantic search and prompt assembly, powering natural-language chat over client DuckDB datasets.
- Built the LLM query layer: business questions to validated SQL over DuckDB, with answers cited back to the source rows.
- Delivered the FastAPI and NestJS services behind the product: query execution, LLM response handling, JWT auth bridge. Built the Next.js dashboard with drag-and-drop layout, KPI cards and auto-generated chart recommendations.
- Hardened the platform for multi-tenant production: PostgreSQL row-level security isolation, Stripe billing, Redis rate limiting and reproducible Docker Compose deploys.

Full Stack Engineer

Mar 2024–Jul 2025

Nashbud Inc

Kolkata, India

- Shipped a MERN e-commerce platform: catalog search and pagination, dynamic-pricing cart, multi-step checkout and order tracking across 3 fulfillment states.
- Integrated Razorpay with HMAC-SHA256 signature verification, idempotency guards and a circuit breaker tripping after 3 failures and self-healing on cooldown. Added a 10-chart admin analytics dashboard.
- Implemented Firebase Auth with role-based access control and 404-cloaked routes. Migrated MongoDB to PostgreSQL 16 via Drizzle ORM using row-level locking to keep writes consistent.

Frontend Developer Intern

Nov 2023–Jan 2024

upliance.ai

Bengaluru, India

- Rewrote CookClub device-control logic with Redux Toolkit and Thunk using an observer pattern.

PROJECTS

News Aggregator API

github.com/man-su-97/news-app-server

Python, FastAPI, PostgreSQL, pgvector, LangGraph, RAGAS, GitHub Actions

- Async FastAPI service ingesting RSS and REST news sources into a unified schema with idempotent PostgreSQL deduplication.
- Hybrid RAG retrieval fusing chunked pgvector semantic search with PostgreSQL full-text search via reciprocal rank fusion, improving Recall@10 over a vector-only baseline with citation-grounded answers.
- LangGraph agent with tool-augmented reasoning, guarded by prompt-injection detection and PII redaction. Retrieval quality is gated in CI on precision, recall, MRR and RAGAS faithfulness scores.

Prism – Multi-Tenant Analytics SaaS

github.com/man-su-97/Prism

Next.js, FastAPI, PostgreSQL, DuckDB, Anthropic Claude API, Redis, Docker

- Multi-tenant SaaS where CSV, Excel and Sheets uploads auto-generate drag-and-drop dashboards of KPI tiles and charts, on a DuckDB and Parquet engine with PostgreSQL row-level security per organization.
- LLM agent running a structured tool-use loop that generates and validates SQL, blocks prompt injection through a validate_sql guard, and streams over SSE with prompt caching on the static schema and system prompt.
- Stripe tiers (Free / Pro / Team) with per-plan LLM token entitlements, Redis rate limiting and revocable email share links.

EDUCATION

B.Tech, Information Technology – CGPA 9.11 / 10

2019–2023

Bengal Institute of Technology

Kolkata, India